

CROSS-SLIT PROJECTION

The Exact Refusal

From a cardboard box to a chain geometry: how a camera with no focal point became a formal system. An essay drawing together the [Tetra engine](#), [Tetra engine complete](#), [Overflow](#) cross-ratio work, and the [Double-slit project](#).

I THE BOX

A cross-slit projection is an empirical fact. It can be demonstrated with a cardboard box, the same way a pinhole camera can: cut, point, expose. The pinhole camera forms its image by passing every projector ray through a single point. The cross-slit camera forms its image by passing every ray through **two orthogonal slits** — in the rig used throughout this work, a vertical slit in one plane and a horizontal slit in another, with the image plane behind both. It is possible to make cross-slit photographs with nothing more than a cross-slit arrangement of cardboard planes.

In mathematics the pinhole is reduced to a point, and the reduction hides a question. No physical pinhole is a point — light could not get through, and it would make no sense for every ray to emerge unscathed from a collision with every other ray. Any working pinhole has a diameter, which invites the question the magnifying glass makes vivid: *how does an image invert at the focus?* (A model of what happens to an image as it passes through the hole: [demo 6](#).)

The cross-slit case carries the same disclaimer. Real slits must have width, else no light. That material fact cannot be ignored, but it can be separated from the underlying geometry and set aside — the way an architecture curriculum separates structures from strength of materials. Everything that follows makes that separation. The physics has done its job the moment the box produces a picture; from here the subject is the math and geometry of cross-slit projection.

II THE DRAFTING TABLE

Cross-slit projection can be interpreted as an internally consistent system of projective geometry, akin to perspective projection. Both live comfortably inside Euclidean space: object, picture plane, and the ambient world coexist as points, lines, and planes on an x-y-z grid, and both project coherent images of three-dimensional objects onto a plane. The classical two-point perspective construction — plan above, elevation beside, sight rays through the station point, projectors dropped to the image area — nests several kinds of space in one drawing: a 3D Euclidean world, a picture plane inside it, and inside *that*, the projected image.

The cross-slit construction is only slightly different. Replace the station point with two slit lines seen edge-on: in the top view the vertical slit appears as a point; in the side view the horizontal slit does. Rays are traced through each in its own view, exactly as in the perspective construction, and the image is assembled on the picture plane. Structure can be imposed on the field itself with a Cartesian grid, so that whole rooms and objects in plan and elevation are cross-slit projected by drafting technique alone ([demo 11](#), [demo 13](#)).

One difference announces itself immediately at the drafting table: the **depth lines bow**. In one-point perspective the lines running into depth are straight spokes through the vanishing point. In the cross-slit image they curve — hyperbolic arcs sagging toward the central vanishing point, with asymptotes parallel to the coordinate grid. The straightness of straight lines is the first casualty of removing the focal point, and it is not a defect; it is the signature. (Formally this is the Bowing Theorem, Theorem 3 of the geometry document, with its three cases explored in [demo 28](#).)

III READING DEPTH

The bowing hints at something the pinhole cannot do: a single cross-slit image carries depth information in the *shapes* of things. A sphere, projected through a pinhole, remains a circle at any distance. Projected through crossed slits, it stretches into an ellipse, and the extent of the stretch is a function of depth. Programs written against the rig geometry can read depth of field directly from the ellipticity of known spheres scattered through the visual field ([demo 14](#), [demo 17](#)).

This is the seed of the codec. If aspect ratio encodes depth, then a flat image of round targets is a volumetric recording, and decoding is geometry rather than inference ([demo 18](#), [demo 19](#), [demo 20](#); overview in the [codec readme](#)). In the calibrated rig — screen at $z = 0$, vertical slit at $Z_v = 150$, horizontal slit at $Z_h = 250$ — the depth-from-aspect-ratio map turns out to be an exact Möbius transformation, which is a fact worth pausing on, because it means the codec's working formula was a piece of projective geometry before anyone recognized it as one. The recognition came later, and it reorganized everything.

IV EUCLID, REVISITED

The original ambition for the formal side of this work was stated plainly: a synthetic geometry — primitives, axioms, theorems — based on Euclidean geometry with a rewritten fifth postulate. Euclid's five are worth reciting because the fifth is the one under pressure. Straight lines join any two points; segments extend; circles exist with any center and radius; right angles are congruent; and — the parallel postulate — through a point not on a line there passes exactly one line parallel to it. The fifth assumes a flat plane.

The cross-slit rig puts a different surface in front of you. The projector rays through the two slits rule a **doubly-ruled surface** — a hyperbolic paraboloid or hyperboloid, depending on section — and on such a surface the parallel question changes character. There is a central ray, perpendicular to every generator that crosses it, plausibly the shortest bridge between all the skew-parallel lines of the ruling. The suspicion recorded in the early notes was that somewhere in this surface the fifth postulate was being renegotiated.

The suspicion was right, but the renegotiation happened somewhere unexpected. The finished synthetic system (*Double-Slit Geometry: Definitions, Axioms, and Theorems*) is not Euclid patched; it is a **chain geometry** in the tradition of Walter Benz — a Minkowski plane, the split-signature sibling of the Möbius plane. And in that system the fate of the fifth postulate is legible. Parallelism does not weaken; it **splits in two**. Two rays are 1-parallel if they share a foot on the first slit, 2-parallel if they share a foot on the second, and these are separate equivalence relations that never confer. Through any ray there is a full pencil of each kind. The structural role Euclid V plays — pinning down exactly one privileged line through an outside point — is played by the Touching Axiom: through a point off a chain

there is exactly *one* chain meeting it at a single prescribed ray. Euclid's parallel survives as tangency.

Where the point went. The pinhole's ray space is a projective plane P^2 ; the cross-slit's is $P^1 \times P^1$ — two projective lines interleaved, a grid rather than a plane, coordinatized not by $\mathbb{R}$ or $\mathbb{C}$ but by the **double numbers** $\mathbb{R} \oplus \mathbb{R}$. The number system of the geometry is two copies of the real line glued along their arithmetic. Explore the axioms interactively in the [geometry companion](#).

V THE HARMONIC THREAD

Perspective and cross-slit projection share one deep invariant: the **harmonic cross-ratio**. In a classical two-point perspective it appears as two pairs of vanishing points arranged harmonically on the horizon line. In a cross-slit construction it appears in terms of the neutral surface and the slits themselves: the depth at which the two slits' magnifications exactly cancel is the harmonic mean of the slit depths, $z^* = 2 \cdot Z_v \cdot Z_h / (Z_v + Z_h) = 187.5$ in the rig — the locus where the four canonical points of a ray sit in harmonic position, cross-ratio -1 , and the image aspect ratio passes through -1 exactly.

The harmonic case is the tip of a larger structure. A cross-ratio λ of four collinear points is not one number but six, depending on the ordering: $\lambda, 1/\lambda, 1-\lambda, 1/(1-\lambda), \lambda/(\lambda-1), (\lambda-1)/\lambda$ — the orbit of λ under the permutation group. Generically all six are distinct (the *anharmonic* case). At the harmonic value the six merge to **three**: $\{-1, 2, 1/2\}$. At the equianharmonic value they merge further to **two**. This collapse — what it looks like, what it means, where it lives — is worked through in the Overflow studies ([the ceiling of the tower](#), [the equianharmonic collapse](#), [modular corner](#)).

An early note in this project reads: *cross-slit projection doesn't fit into this framework*. That turned out to be exactly half right. It does not fit because the framework assumes a single number, and the fundamental invariant of cross-slit geometry is a **pair**: the bi-cross-ratio (λ, μ) , one cross-ratio per slit, equivalently one cross-ratio valued in the double numbers. The six-element orbit acts *diagonally* on the pair, which doubles the classical taxonomy into configurations with no pinhole analogue: harmonic on one slit and generic on the other (*split-harmonic*), harmonic on both (*bi-harmonic*, $(-1, -1)$), harmonic on one and

equianharmonic on the other. The collapse from six to three now happens per axis, and the neutral surface is its physical address — the one depth where the rig itself sits at the harmonic value on both slits at once. The framework did not exclude cross-slit projection; it was one factor of it.

VI THE SEAM

With cross-ratios, the point is the primitive: four points on a line, one number. What is the primitive of cross-slit geometry? Not the point. Two generic rays of the system meet *nowhere* — the only meetings allowed are between parallel rays, and those meetings are exiled to the slits themselves. The union of the two slits, belonging to every ray's addressing but to no ray's interior, is the **seam**. The seam between two systems is the atom, and the entire geometry is built out of it.

In this work the interface-belonging-to-neither is a locus arrived at, found after the fact as the excluded seam. That inversion may be why cross-slit has always felt like home ground rather than a result reached from elsewhere. What the seam *is* gets no assertion here — establishing that is precisely the project, and the companion essay [The Seam](#) is where the establishing happens: algebraically the seam is the zero-divisor locus of the double numbers, and since $\mathbb{R} \oplus \mathbb{R}$ is the split-complex plane of 1+1 relativity, two rays that meet in space are null-separated in the image. The seam is the picture's light cone.

This non-concurrence isn't a weaker pinhole. It is the exact refusal of the one thing that makes a pinhole a pinhole.

And the refusal is nearly unique. Kummer's classical study of congruences of order one says that if every scene point is to be threaded by exactly one ray, the aperture has very few options: a point, two skew lines with their degenerations, or the secants of a twisted cubic. The camera built here is not one arbitrary alternative among many; it is nearly the only alternative geometry permits.

VII DOUBLING THE RIG

One page of the early notes proposes a way to *read* depth rather than compute it: take two cross-slit projections of the same scene at once, rotated 90° to each other about the viewing axis, separated by filters — red and blue, or polarized — and read the disparity not as positional offset, as in ordinary stereo, but as **relative shape distortion**. The greater the separation between related pairs of red and blue features, the closer the feature.

That sketch became the dual cross-slit rig. Worked out formally, the second projection is a second tetrahedron rotated 90° about the shared axis; the combined aperture at each slit plane is a cross; and the channel-to-arm assignment solves image registration by construction. The zero-disparity surface — the shape-horopter, where the two channels fuse — is the neutral surface again, $z^* = 187.5$: the harmonic locus of section V, now doing optical work. The dual ratio cancels ellipticity errors that fool a single channel, exactly and algebraically, and two independent channels buy the expected $\sqrt{2}$ in signal-to-noise. The idea that began as a colored-filter sketch closes the loop with the oldest invariant in the project.

VIII PROVENANCE, AND WHAT IS CLAIMED

Honesty about provenance. The underlying objects are classical. The family of transversals to two skew lines is the hyperbolic linear congruence, studied since Plücker, Kummer, and Sturm in the nineteenth century. The camera model appears in the modern imaging literature as the crossed-slits (X-slit) projection of Zomet, Feldman, Peleg, and Weinshall (2003), and as a class within the General Linear Camera taxonomy of Yu and McMillan (2004). The abstract axiomatics instantiates the theory of chain geometries and Minkowski planes developed by the school of Walter Benz.

What this body of work contributes is the **assembly**: the Erlangen-style presentation of the camera as a geometry in its own right; the identification of the working codec formula as a cross-ratio, which explains both its Möbius form and its noise behavior; the drafting-table constructions; the interactive demonstrations that verify every numbered theorem against the live rig; and the dual-rig development. Speculative material stays quarantined from the defensible claims, and the demos carry their own check scripts. The boxes were cardboard;

the geometry, it turns out, was always chain.

Companion pieces: [Double-Slit Geometry: Definitions, Axioms, and Theorems](#) · [The Seam](#) · [Interactive companion](#) · [Addenda](#). Rig constants throughout: screen $z = 0$, vertical slit $Z_v = 150$, horizontal slit $Z_h = 250$, $D = 100$.